AarogyaTwin-X architecture

- Local synthetic research prototype
- Forecast heads at 30, 60, and 120 minutes
- Evidence, uncertainty, provenance, and abstention are first-class outputs

End-to-end data flow

- Seed 2026 -> synthetic simulator -> patient-disjoint split
- Past-only feature builder -> CPU temporal model
- Calibration and bounded residual -> joblib/JSON artifacts
- FastAPI local API -> React/Vite dashboard

Synthetic data boundary

- 60 patients, 5 days, 5-minute grid
- CGM, heart rate, HRV, sleep, activity, meals, and static fields
- Hidden simulator parameters never become model features
- No real patient records or public benchmark data

Leakage-aware evaluation

- 36 train patients; 6 calibration-tuning; 6 calibration-interval; 12 test
- No test patient tunes models, calibration, thresholds, or intervals
- Every feature vector is bounded by its replay anchor
- Correlated replay windows are not independent subjects

Model layers

- Temporal lag gradient-boosting heads for glucose
- Fusion logistic classifier for the sustained-event endpoint
- Train-only imputation, standardization, and missing indicators
- Platt probability calibration and split-conformal intervals

Causal-in-time residual

- A prior forecast error is used only after its target timestamp is observable
- Per-patient, short, bounded error history
- Strength selected on calibration-tuning patients
- Adaptive residual only - not a clinical insulin-sensitivity model

Serving boundary

- TwinEngine loads artifacts and never trains during API requests
- Local endpoints: health, patients, twin snapshot, scenario, metrics
- Missing/stale CGM produces warnings and abstention
- No fallback medical numbers are inserted

Dashboard boundary

- Replay held-out patient history
- Three named forecast heads and intervals
- Event state and risk when eligible
- Dropout, provenance, limitations, and assumption-based scenario view

Non-claims

- No clinical accuracy or real-patient generalization
- No causal treatment effect or dosing recommendation
- No arbitrary distribution-shift robustness
- Full saved evidence is in docs/EVIDENCE_SUMMARY.md